

Abstract No.

Patient-derived glioblastoma tumoroids preserve tumor identity and reveal neurotropic stem-cell behavior

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Glioblastoma (GBM, IDH-wild type) is an aggressive primary brain cancer driven by glioblastoma stem cells (GSCs), which invade the brain through a “wired network” of microtubes and establish GSC–neuron synapses. How GSCs dynamically establish these interactions remains unknown. Here, we establish optimized methods to generate 3D GBM tumoroids that faithfully preserve the molecular features of parental tumors and enable long-term, high-resolution imaging of tumor–brain interactions. Continuous spatiotemporal tracking revealed striking neurotropic behavior, with GBM tumoroids releasing asymmetric, vesicle-like particles that assemble into Milky Way-like migratory streams extending toward brain organoids. These streams guide directional GSC migration and facilitate GSC–neuron synaptic interactions, revealing a previously unrecognized mode of GBM–brain communication. Thus, our study, revealing a vesicle-associated migratory stream emerging from GBM tumoroids, may form a complementary “wireless network” for GBM–neuron communication. With a success rate exceeding 90%, these cryopreservable tumoroids could enable translational GBM modeling.



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